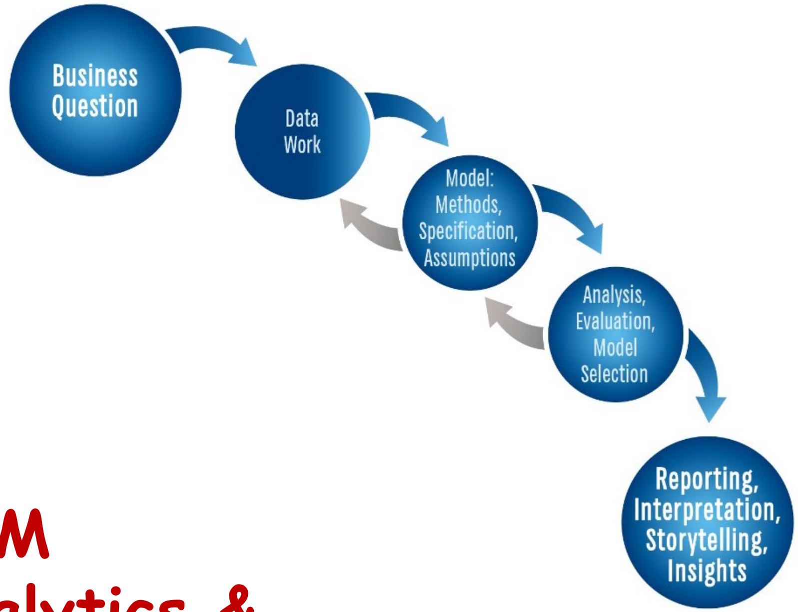

PAML4M
Predictive Analytics & Machine Learning for Managers
7. Dimensionality

Prof. J. Alberto Espinosa

Last updated 8/1/2026

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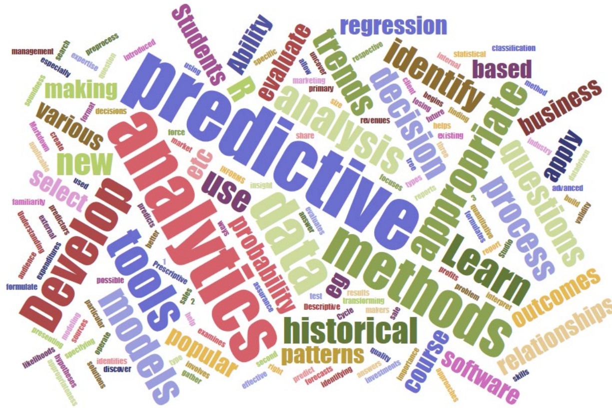

ITEC 621
Predictive Analytics
7.1 Ridge & LASSO

Use PAML4M_7RA_Dimensionality_Ridge&LASSO.Qmd

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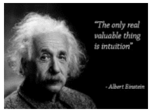
Regularization (Penalized or Shrinkage) Models

XI(×) – X's are not independent (are correlated)

Addressing Dimensionality



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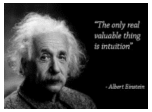
Addressing Dimensionality: Intuition

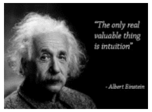
- Sometimes dimensionality issues can be resolved through variable selection → if predictors can be added or removed as needed and multi-collinearity is at tolerable levels with the variables selected, then dimensionality is taken care of.
- But we often need to retain all or most predictors (for business reasons or otherwise) and dimensionality issues (e.g., high variance and multi-collinearity) makes the model unstable.
- For example, molecular, biology and some business regression models often require hundreds or thousands of predictors that cannot be removed, making them ideal for shrinkage models
- One way to address dimensionality without dropping variables from the model is by shrinking the model coefficients
- Shrinkage introduces bias but also reduces variance → a tradeoff !!



Tuning Parameters

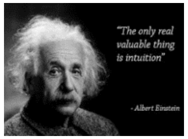
- A **tuning parameter** in a model is something we can change to refine and improve a model
- Most models that allow for tuning parameters employ the symbol λ to denote them
- **Examples** of tuning parameters:
  - ➢ How large to grow a **tree**?
  - ➢ In a classification model, what is the **probability threshold** (40% 50%, 60% ?) to classify an observation as a **1** or **0**?
  - ➢ In **regularization** models, **how much** we **shrink** coefficients?
  - ➢ In **dimension reduction**, reducing **P** dimensions to **M** (P >> M)
- In most modeling methods, the **optimal** tuning parameter λ for a particular model is the one that provides the **best cross-validation test error**

Regularized, Penalized or Shrinkage Models



Regularized, Penalized or Shrinkage Methods

- These 3 labels mean the same thing – use them interchangeably
- These methods aim at shrinking (and thus biasing) the regression coefficients to avoid issues of multi-collinearity
- But shrinkage also reduces the model variance in the presence of multi-collinearity, providing more accurate and reliable predictions.
- There are many regularization methods. We explore two of them here:
  - **Ridge Regression**
  - **LASSO**
  - **Note:** an **Elastic Net** model uses a weighted average of Ridge and LASSO methods



Shrinkage Methods: Intuition

- With variable selection methods discussed earlier, the main decisions are about which predictors to include and/or remove predictors based on F tests, MSE, and other fit statistics
- Removing a variable from a model is equivalent to forcing its β coefficient to 0
- In contrast, "penalized" (or "shrinkage") methods fit a model with all P predictors but shrink (or "regularize") the coefficient estimates towards 0, but not always 0.

How Much to Shrink?

- With variable selection methods the final model is somewhere in between the Null (no predictors) and the Full model, and we add and/or remove variables until we find an optimal set of predictors to retain in the model.
- With shrinkage methods, the final model is also somewhere in between the Null model and the Full model. But rather than removing variables, we start with the Full model and shrink all the coefficients by a factor λ
- λ is a tuning parameter ranging from 0 (no shrinkage – i.e., Full model) to ∞ (all coefficients shrunk to 0 – i.e., Null model).
- The main goal is to find the shrinkage factor λ that yields the optimal model with the best CV Test results.
- That is, the final model will include all predictors, but their weight in the prediction will be lowered by the shrinkage factor λ.

Variable Selection vs. Shrinkage

Full Model *(**ALL** predictors):* $Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \ldots\ldots + \beta_P(X_P) + \varepsilon$

Variable Selection

$\beta_1(X_1) + \beta_3(X_3) + etc.$

*Some predictors are **retained***

*Some other are **dropped***

Shrinkage Methods

$\beta_{1S}(X_1) + \beta_{2S}(X_2) + \ldots\ldots + \beta_{PS}(X_P)$

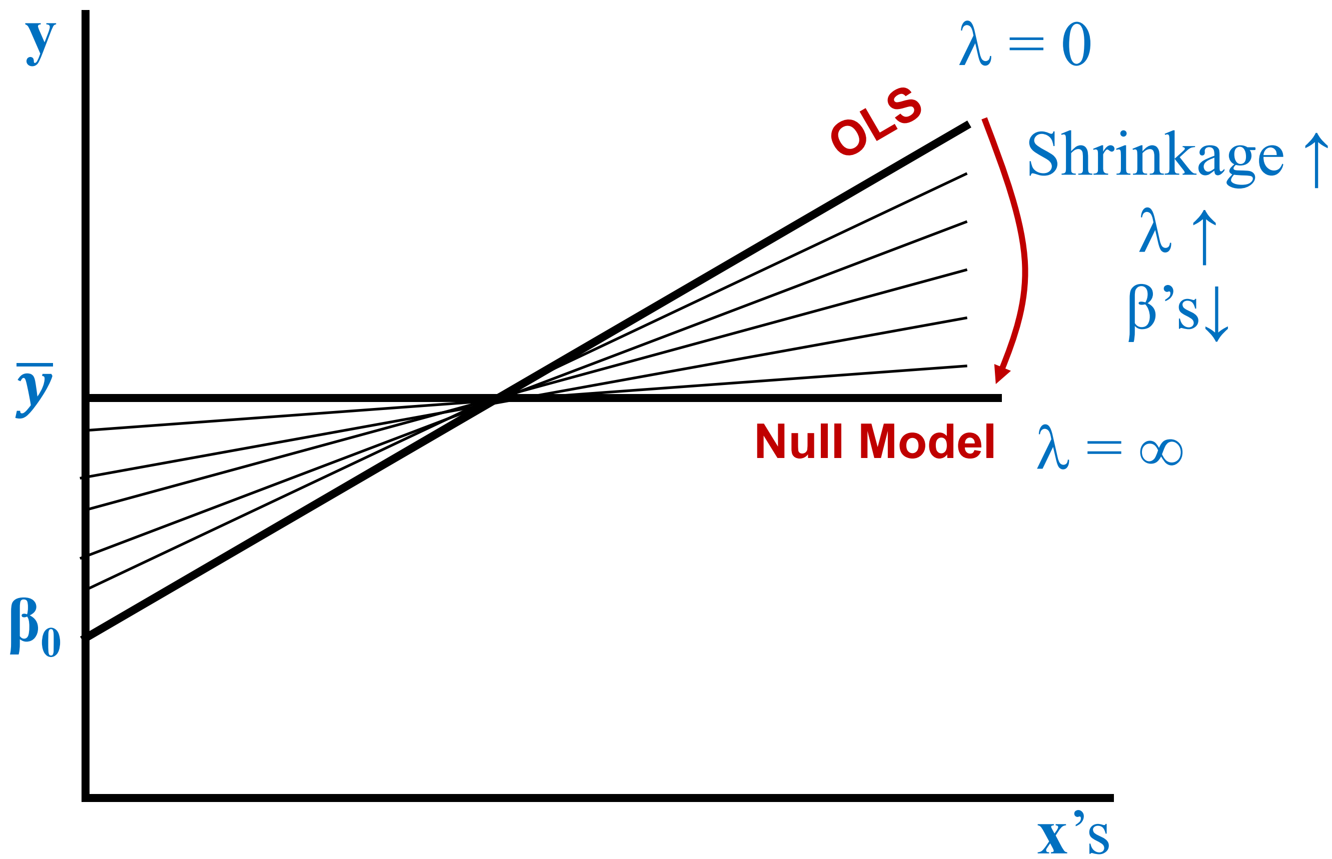
Where $\beta_{1S} <= \beta_1$*, etc. (i.e., shrunk)*

*All predictors are **retained** but:*

- ✓ β_S *are the **shrunk** coefficients*
- ✓ *X's are **standardized***
- ✓ *But* β_S *are **unstandardized** back to raw*

Null Model *(**NO** predictors):* $Y = \beta_0 + \varepsilon$

Shrinkage in Action



Ridge and LASSO Regressions

Ridge and LASSO Regressions

- Most common penalized (shrinkage or regularization methods are Ridge and LASSO regressions
- All predictors that matter for business reasons are included, but predictors are shrunk (i.e., penalized, rather than removed) so that small coefficients have a very low weight in the prediction, but not removed.
- Shrinking regression coefficients increases the **bias** of the model (because they are no longer the true coefficients, but are artificially shrunk), but shrinkage often reduces variance and improves predictive accuracy compared to variable selection methods.
- This is particularly true for models with high dimensionality

- How much **shrinkage** is controlled with a **“tuning”** parameter λ

Ridge Regression

XI(×) – X's are not independent
(i.e., are highly correlated → high dimensionality)

Ridge Regression

- OLS finds regression coefficients that minimize the SSE
- Ridge regression minimizes the SSE + a shrinkage penalty:

$$SSE_R = SSE + \text{shrinkage penalty} = SSE + \lambda\,\underbrace{(\beta_{S1}^2 + \beta_{S2}^2 + \beta_{S3}^2 + etc.)}_{L2\ Norm^2}$$

Technical Notes:

1. All variables, y and x's, are standardized before fitting the Ridge regression. Otherwise, the scale of y and x's would have a huge influence on the resulting coefficients.
2. But the β_S coefficients are un-standardized back to their normal scales when reported.
3. λ is the shrinkage (tuning) parameter. Normally, Ridge reports results with **100** different λ's. We can then retrieve coefficients for any of these λ's and find the best λ that minimizes the CV MSE.

Ridge Regression: Intuition

- In sum,
  - OLS minimizes SSE → Ridge minimizes SSE plus a penalty
  - We can vary λ thus controlling the shrinkage
    - ✓ If we set $\lambda = 0$, Ridge minimizes SSE → same as OLS →
    - ✓ If we set $\lambda = \infty$, Ridge yields the null model $y = \beta_0$
    - ✓ If we set λ = large, the resulting β's will be small and vice versa
  - Small λ → large coefficients → less bias, more variance
  - High λ → small coefficients → more bias, less variance
- **Notes:**
  - **Ridge doesn't shrink coefficients to 0, except when $\lambda = \infty$**
  - The coefficients simply become really small with large values of λ thus having little influence on predictions
  - In contrast, the LASSO method does shrink small coefficients all the way to 0 with large values of λ

How to use Ridge Regression

- Ridge regression is particularly useful for large models with lots of predictors and dimensionality issues (e.g., multi-collinearity)
- Ridge regression coefficients are more biased but have less variance than OLS
- OLS is "scale invariant" → re-scaling a variable (e.g., feet to meters, dollars to thousands), the OLS coefficients will change proportionally.
- This is NOT the case with Ridge → The penalty in Ridge regression is based on the sum of the squared coefficients → re-scaling changes the results disproportionately
- Therefore, it is standard practice to:
  - ✓ **Standardize** the predictors in Ridge regression models, then unstandardized the resulting Ridge coefficients to raw values.
  - ✓ **Compare** → try several λ's and select the one with the best cross-validation test results

L2 Norm

- $l2\ Norm$ is a popular measure of the actual overall shrinkage of the Ridge regression coefficients
- Ridge's Shrinkage penalty $= \lambda\ \underbrace{(\beta_{S1}^2 + \beta_{S2}^2 + \beta_{S3}^2 + etc.)}_{l2\ Norm^2} = \lambda * l2\ Norm^2$

  - $l2\ Norm = \sqrt{\beta_{S1}^2 + \beta_{S2}^2 + \beta_{S3}^2\ + etc.}$
- As λ grows
  - All coefficients shrink proportionally
  - $l2\ Norm$ gets smaller providing a measure of total shrinkage

Ridge Regression

- {libary} Topic
  `Search keywords`
- Ridge Regression
  ```
  library(glmnet)
  glmnet(x, y,
  L2 Norm
  Optimal Shrinkage with CV Testing
  Extracting Ridge Coefficients
  Predicting with Ridge
  ```

LASSO Regression

XI(×) – X's are not independent
(i.e., are highly correlated → high dimensionality)

LASSO Regression: Intuition

- **Same** as **Ridge**:
  - OLS minimizes SSE → LASSO minimizes SSE plus a penalty
  - We can vary λ thus controlling the shrinkage
    - ✓ If we set λ = 0, Ridge minimizes SSE → same as OLS →
    - ✓ If we set λ = ∞, Ridge yields the null model $y = \beta_0$
    - ✓ If we set λ = large, the resulting β's will be small and vice versa
  - Small λ → large coefficients → less bias, more variance
  - High λ → small coefficients → more bias, less variance
- Different from Ridge:
  - **In contrast to Ridge, LASSO shrinks small coefficients (squashes them) to 0 with large values of λ**
- For this reason, LASSO falls between Variable Selection and Ridge
- In fact, LASSO can be used as a variable selection method → drop variables whose coefficients shrink to 0, and then try OLS

LASSO Regression

(Least Absolute Shrinkage and Selection Operator)

- Again, OLS finds regression coefficients that minimize the SSE:
- LASSO regression finds coefficients that minimize:

$$SSE_L = SSE + \text{Shrinkage Penalty} = SSE + \lambda\ \underbrace{(|\beta_{S1}| + |\beta_{S2}| + |\beta_{S3}| + etc.)}_{L1\ Norm}$$

- That is, the penalty λ is applied over the sum of the absolute values of the coefficients, rather than over the sum of their squared values

Technical Notes:

1. Like Ridge, all variables, y and x's, are standardized in LASSO
2. Also, the β_S coefficients are un-standardized back to their normal scales when reported.
3. λ is the shrinkage (tuning) parameter, just like in Ridge

LASSO Regression: More Intuition

- The effect is similar to Ridge regression:
  - $\lambda = 0$ → minimizes SSE → LASSO = OLS
  - $\lambda = \infty$ → LASSO = null model
- One important and interesting difference → mathematically, Ridge coefficients can never be shrunk to 0 (except when $\lambda = \infty$)
- But some LASSO coefficients do eventually become exactly 0 as λ increases

  → LASSO falls in between Variable Selection and Ridge
- In fact, some analysts use LASSO as a variable selection method, dropping all the variables whose coefficients shrink to 0, and then try OLS or other methods with the variables that remain

How to use LASSO

- LASSO and Ridge regression are very similar, compared to OLS → more bias, less variance, affected by scale variant, etc.
- Therefore, like with Ridge, it is standard practice to:
  - ✓ **Standardize** the predictors in LASSO regression models, then unstandardized the resulting Ridge coefficients to raw values
  - ✓ **Compare** try several λ's and select the one with the best cross-validation test results
- Because LASSO shrinks small coefficients to 0 → LASSO is a preferred method if it is not important to retain all available variables in the model, otherwise use Ridge regression
- Like with Ridge, it is standard practice with LASSO to try several λ's and select the one with the best CV Test results

L1 Norm

- $l1\ Norm$ is a popular measure of the actual overall shrinkage of the LASSO regression coefficients

- LASSO's Shrinkage penalty = $\lambda\ (\underbrace{|\beta_{S1}| + |\beta_{S2}| + |\beta_{S3}| + etc.}_{l1\ Norm}) = \lambda * l1\ Norm$

$$l1\ Norm = |\beta_{S1}| + |\beta_{S2}| + |\beta_{S3}| + etc.$$

- As λ grows
  - All coefficients shrink proportionally
  - $l1\ Norm$ gets smaller providing a measure of total shrinkage

LASSO Regression

- {libary} Topic
  `Search keywords`
- LASSO Regression
  ```
  library(glmnet)
  glmnet(x, y,
  L2 Norm
  Optimal LASSO Lambda
  Extracting LASSO Coefficients
  Predicting with LASSO
  ```

Quick Notes on Shrinkage:

- If your analytics goal is predictive accuracy, select the λ shrinkage value that yields the lowest CV RMSE, misclassification or deviance

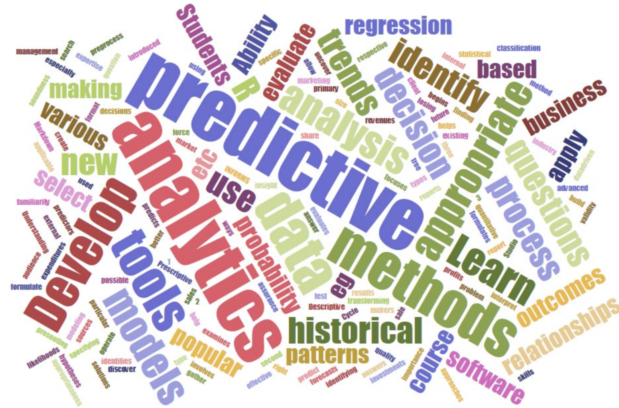
- If your analytics goal is interpretation or inference, you want to have the least biased coefficients possible. If so, select a relatively small λ shrinkage value that yields an acceptable CV RMSE

Elastic Net Regression

Final Note About α: Elastic Net Regression

- The Ridge syntax is ridge.mod <− glmnet(x, y, alpha = 0)
- The LASSO syntax is lasso.mod <− glmnet(x, y, alpha = 1)
- So, what is this α (alpha)?
- The authors of the {glmnet} package (and of the ISLR text), programmed a single glmnet() function to fit both models →

  Minimize SSE + (1- α) Ridge penalty + (α) LASSO penalty
- So, conveniently and cleverly, when:
- α = 0 → glmnet() fits a Ridge regression
- α = 1 → glmnet() fits a LASSO regression
- 0 < α < 1 → glmnet() fits an Elastic Net regression = some Ridge + some LASSO



ITEC 621
Predictive Analytics
7.2 PCR and PLS

Use PAML4M_7RB_Dimensionality_PCR&PLSR.Qmd

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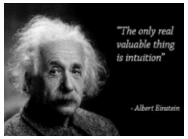
Dimension Reduction Models

XI(×) – X's are not independent
(i.e., are highly correlated → high dimensionality)

Dimension Reduction: Overview

- Some models may need to include many important predictors, which is problematic when: (1) the ratio of predictors to observations is large (low degrees of freedom) and when (2) predictors are correlated (i.e., co-linear)
- The methods covered so far have addressed dimensionality by either using a subset of variables or by shrinking coefficients.
- This may not be feasible or practical with models that require hundreds or even thousands of predictors (e.g., biology, surveys).
- In such cases, it helps to explore the linear relationships among predictors and use the observed correlation to create new variables as linear combinations (i.e., components) of the original variables.
- When we do this, a few of the new components may explain a large portion of the variance in the data, thus helping reduce the model dimension without losing much predictive power.



Dimension Reduction: Intuition

- If we have P somewhat correlated predictors, we can combine them into M components, with M << P → fewer variables
- The idea is to use the correlation matrix to figure out how to group predictors into group called components
- Dimension reduction: is about reducing the estimation of P+1 coefficients (β_0, β_1, β_2,… β_P) to M+1 coefficients (α_0, α_1, α_2,… α_M)

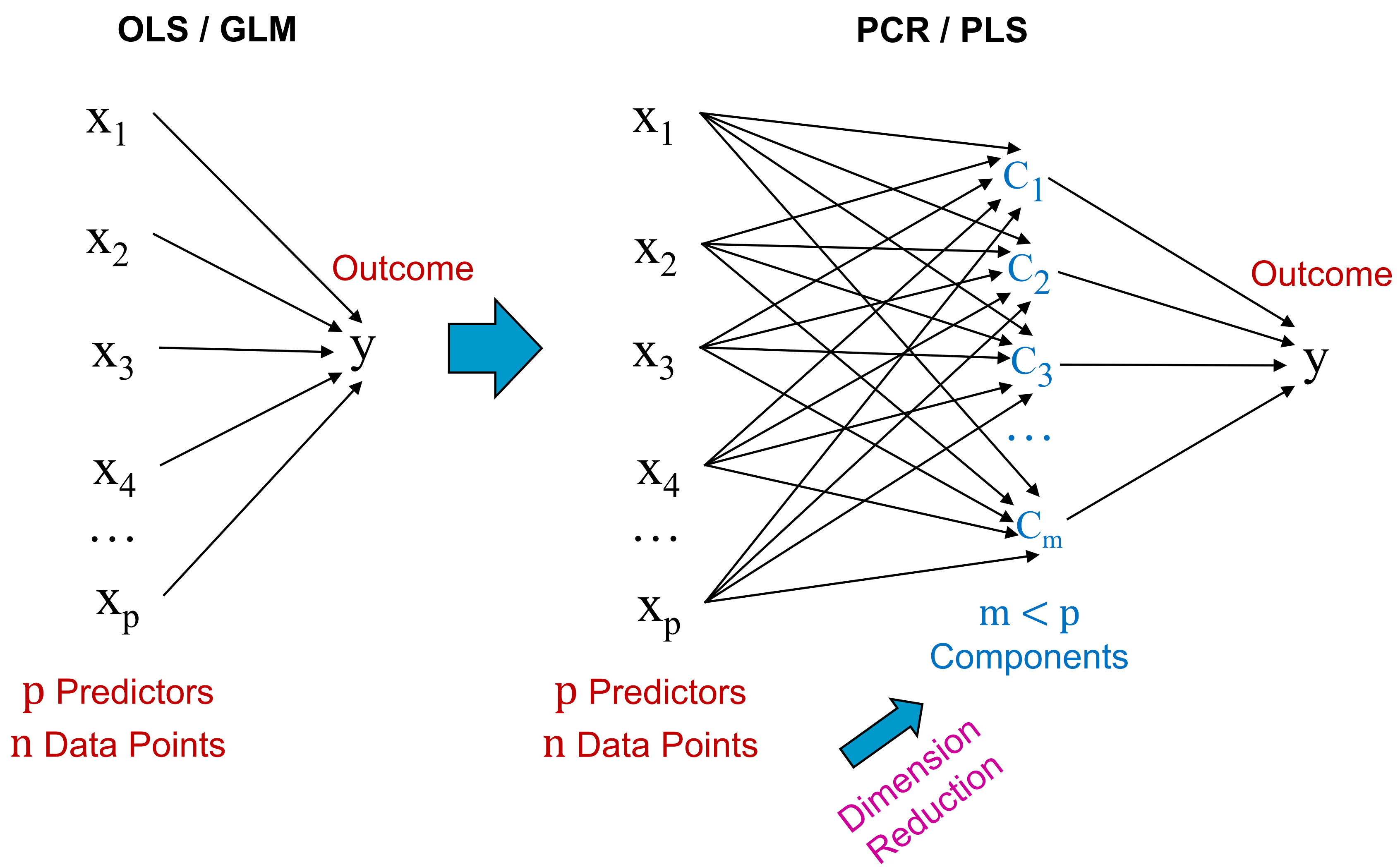
  Example: if a vehicle's volume, horsepower, and weight are negatively correlated with its gas mileage, but we know that these 3 predictors are highly correlated with each other, we can take advantage of this correlation and combine them into a new variable called something like "car size" composed of some percentage of volume, plus some of horsepower, plus some of weight, reducing the model variables from 3 to 1.

- We may gain predictive accuracy at the expense of some interpretability → so, it is a tradeoff !!

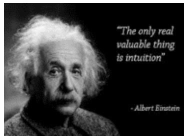
Dimension Reduction Methods

- Two popular dimension reduction methods are:
  - **Principal Components Regression** (**PCR**) and
  - **Partial Least Squares** (**PLS**)
  - Both methods use the correlation matrix of P predictors to find M (<P) linear combinations of the P predictors
- These methods may increase bias but substantially reduce variance of the coefficients, particularly when P is large relative to N

Dimension Reduction: Illustration



Principal Components (PCs)



Principal Components (PCs): Intuition

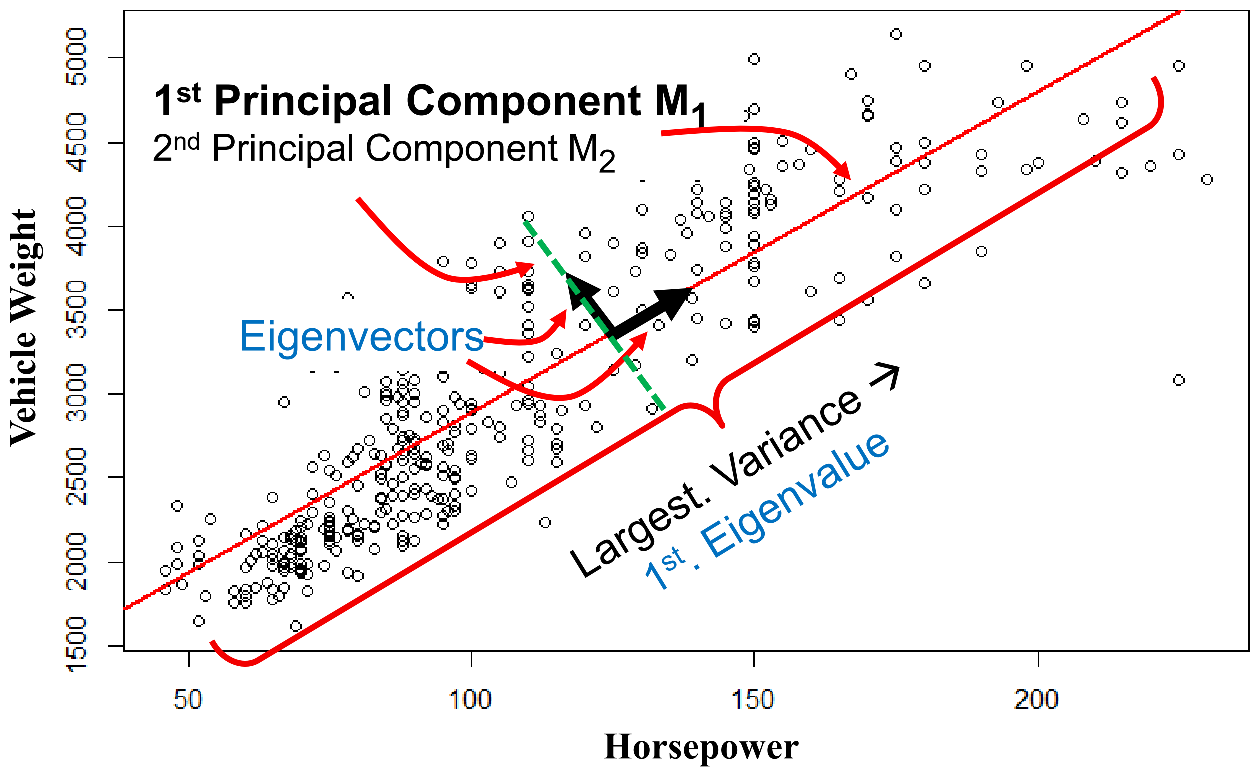
- If you have P predictors, you can identify the linear combination of these predictors that yields the highest variance
- This linear combination is called the 1st Principal Component (1st PC)
- We then evaluate all other linear combinations that are perpendicular (i.e., "orthogonal") to the 1st PC and find the one with the next highest variance, which is the 2nd PC; and so on.
- A model with P predictors has exactly P Principal Components
- Some very useful properties of these PCs:
  - ✓ The PCs are perpendicular to (i.e., "orthogonal" to or "independent" from) each other → 0 correlation → no multicollinearity)
  - ✓ They are sorted from highest to lowest variance dimensions
  - ✓ If the predictors are highly correlated, the first few M PCs (M << P) will account for a large proportion of the variance in the predictors.
  - ✓ Allowing us to reduce the model dimension from P variables to M components

Principal Components: Illustration

This can be illustrated with 2 variables like vehicle Horsepower and Weight (as predictors of Gas Mileage), resulting in 2 PCs

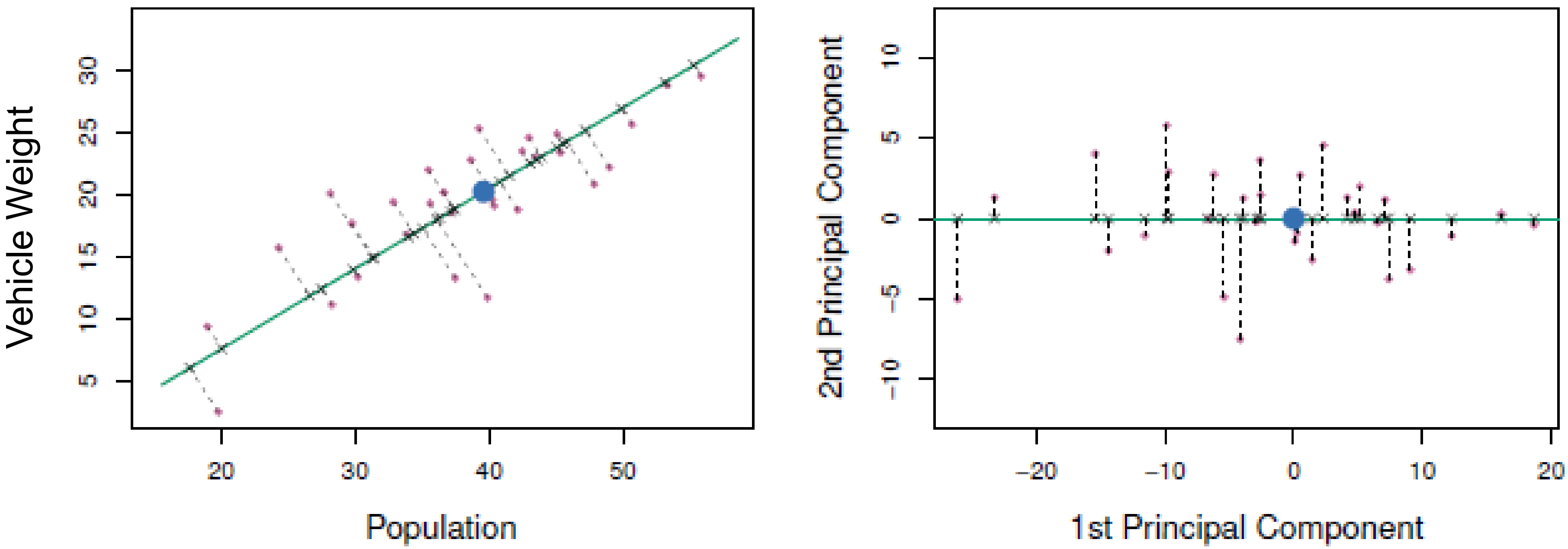
Think of PCs as axes rotated into the direction of highest variance and centered around the mean

It is clear from this plot that Horsepower and Vehicle Weight are highly correlated. But the PC's M_1 and M_2 are not



Orthogonality

Once we rotate the 2 PCs it becomes clear that the two perpendicular components have 0 correlation – i.e., they are "independent" or "orthogonal"



Principal Components Explained

- In the previous plot, the First PC is actually:

$$\mathrm{M1} = 0.839\,(\mathrm{hp} - \overline{\mathrm{hp}}) + 0.544\,(\mathrm{wgt} - \overline{\mathrm{wgt}})$$

- Notice that the PCs are centered at the means of hp and wgt
- 0.839 and 0.544 called the Loadings of PC $\mathrm{M1}$ → they are the weights of the pop and ad variables, respectively, which establish the direction of M_1
- Notice an important property → $0.839^2 + 0.544^2 = 1$
- More generally:
  - For P variables $X_1, X_2, \ldots\ldots, XP$,
    There are P principal components $M_1, M_2, \ldots\ldots, MP$
  - Each PC has P loadings, 1 for each X variable
  - The loadings are scaled so that the sum of the squared loadings for each component = 1
  - All components are orthogonal (i.e., independent)

Components, Loadings & Scores

- For a set of P predictors → $x_1, x_2, \dots, x_P$
- There are P orthogonal (i.e., perpendicular, uncorrelated) PCs:

$$M_1 = l_{11}x_1 + l_{12}x_2 + \cdots + l_{1P}x_P$$
$$M_2 = l_{21}x_1 + l_{22}x_2 + \cdots + l_{2P}x_P$$
$$\dots.$$
$$M_P = l_{P1}x_1 + l_{P2}x_2 + \cdots + l_{PP}x_P$$

- The sum of squared loadings for each component $= 1$
- The sum of squared loadings for each predictor $= 1$

- For a data point i with predictors $x_{i1}, x_{i2}, \dots\dots x_{ip}$ the corresponding PC values $M_{i1}, M_{i2},$ etc. are called the "PC Scores" for that data point

Principal Components Regression (PCR)

Principal Components Regression (PCR)

- The main purpose of PCR is to reduce the dimensionality (i.e., number of variables) of a model without removing predictors.
- Since the PC's are sorted from highest to lowest variance, it is possible that the first few PC's may explain a high portion of the variance in the data.
- So, for a given OLS regression with P predictors:

$$Y = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \ldots\ldots + \beta_P(X_P) + \varepsilon$$

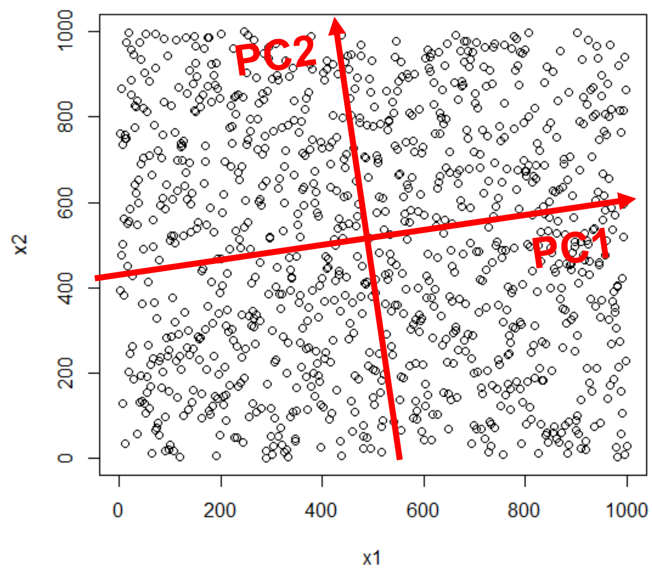
- We can construct another OLS model on M PC's:

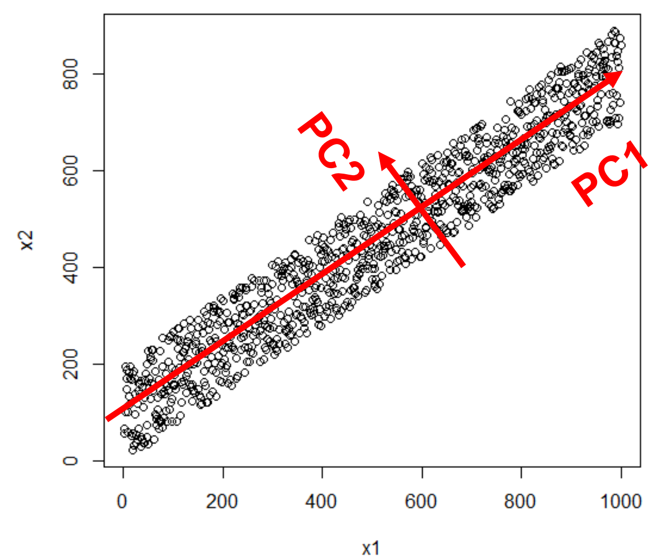
$$Y = \alpha_0 + \alpha_1(M_1) + \alpha_2(M_2) + \ldots\ldots + \alpha_M(M_M) + \varepsilon$$

- M can be anywhere between 1 and P PC's
- If M << P → substantial dimension reduction; otherwise, not useful
- Since each PC is a linear transformation of all predictors → all P predictors are represented
- The key is to find the optimal number of M PC's (i.e., a tuning parameter) to include in the model
- As with regular OLS, as M ↑ → Bias ↓ and Variance ↑

When is PCR Beneficial?

1. Good for prediction ☺; Less so for interpretation ☹
2. When dimensionality is high →
   - ✓ There is a low ratio of observations to predictors
   - ✓ There is high multi-collinearity among predictors
3. PCR is beneficial if the first few PC's explain a sufficiently large proportion of the variance in the data. For example, with 2 predictors:



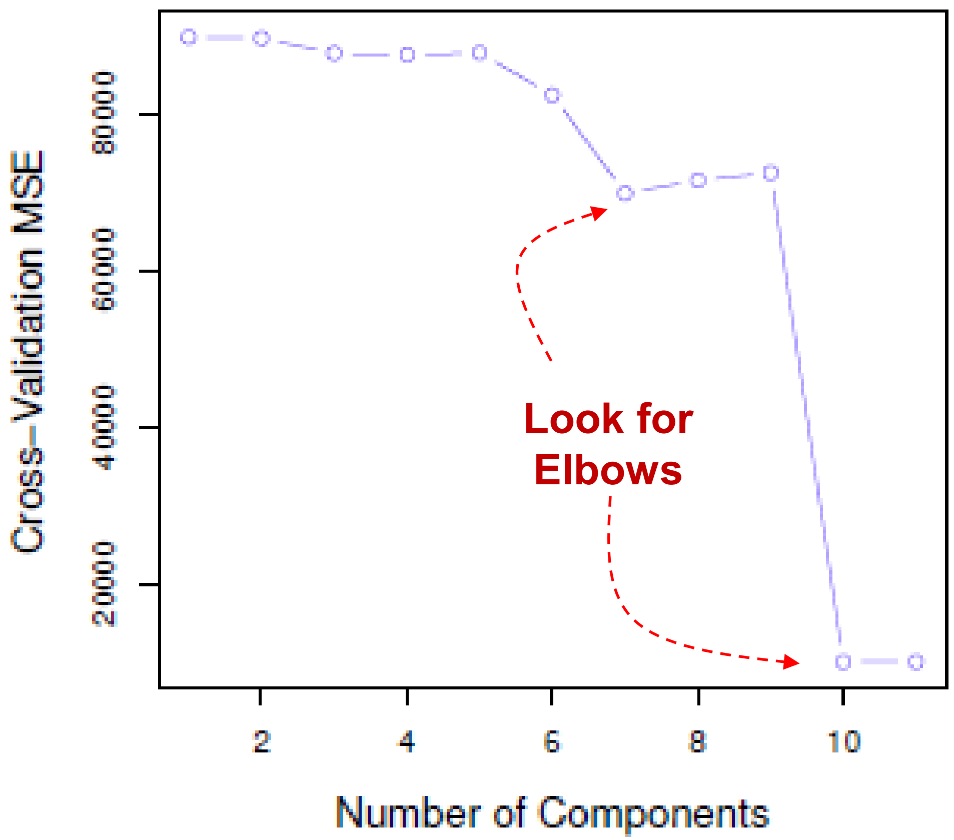


Predictors x1 and x2 are uncorrelated
→ PCR is not very useful !

x1 and x2 are correlated → PC1 explains most of the variance in the data
→ PCR is beneficial !!

Choosing M (the Number of PC's)

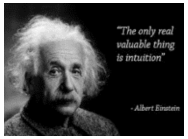
- P predictors have P PC's. How many do we use?
- As with other methods, CV Testing is the best selector for M
- In this illustration using the Credit data set, we see that the 10 K-Fold MSE drops sharply from 9 to 10 PC's and then stays flat.



Principal Components Regression (PCR)

- {libary} Topic
  ```
  Search keywords
  ```
- LASSO Regression
  ```
  library(pls)
  pcr.fit <- pcr(Salary ~ .,
  validationplot(pcr.fit,
  PCR Coefficients
  Predictions with PCR
  ```

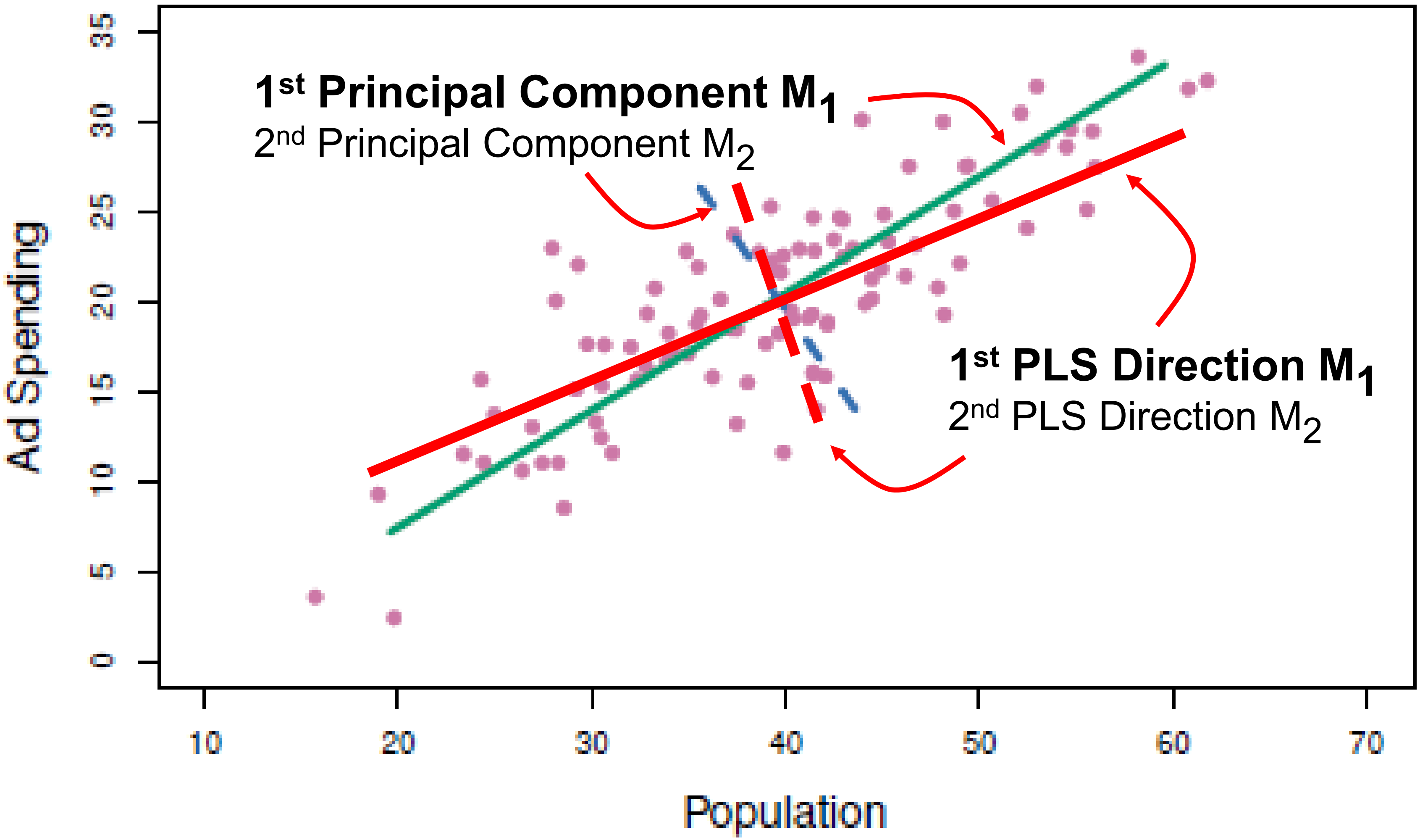
Partial Least Squares (PLS) Regression



Partial Least Squares (PLS): Intuition

- With PCR, the $x_1, x_2, \ldots\ldots x_p$ variables are transformed into $M_1, M_2, \ldots\ldots M_p$ components in an unsupervised way
- That is, the predictor variable dimensions are rotated to find directions in which the data exhibits highest variance.
- This is an unsupervised method because the outcome variable Y is not taken into account when doing PCA
- While PCR does well in general, there is no guarantee the first few M components will be the best directions to predict Y
- In contrast, PLS is a supervised method
- Like PCR, PLS is a dimension reduction method, but unlike PCR, PLS does further rotation of the PC's to maximize their correlation of with the outcome variable Y.
- In sum, PLS attempts to find directions that not only explain the predictors but also the outcome variable
- It does this by placing stronger weight on variables that are more strongly correlated with Y

PLS: Illustration



Components M_1 and M_2 are no longer PC's but are close

PLS Regression

- In PLS the components are no longer PC's because they deviate from the directions of higher variance, so they are simply called first, second, etc. "directions".
- Like with PCR, PLS starts with OLS (X's are standardized):

$$Y = \beta_0 + \beta_1(x_1) + \beta_2(x_2) + \ldots\ldots + \beta_P(x_P) + \varepsilon$$

- And finds M "directions" that are linear combinations of the X's:

$$Y = \alpha_0 + \alpha_1(M_1) + \alpha_2(M_2) + \ldots\ldots + \alpha_M(M_M) + \varepsilon$$

- M can be anywhere between 1 and P (all PC's → same as OLS)
- But the identification of the first direction M_1 starts by placing more weight on the variable that has the strongest correlation with Y
- Like with PCR, M is a tuning paramerer.
- As with OLS and PCR, in PLS as M ↑ → Bias ↓ and Variance ↑
- Because PLS is a <u>supervised</u> method, the coefficients are often less biased but have more variance than PCR
- Again, cross-validation is the best selector to find the optimal M and to compare PLS with other models including PCR

Partial Least Squares (PLS) Regression)

- {libary} Topic
  `Search keywords`

- Partial Least Squares (PLS) Regression
  ```
  library(pls)
  pls.fit <- plsr(Salary ~ .,
  validationplot(pls.fit,
  PLSR Coefficients
  Predictions with PLSR
  ```

PLSR vs. PCR Final Note

Important Technical Point: PLSR does further rotation of the PCR components to improve the correlation with the outcome variable. But it only rotates the components in each model. For example, in the PCR model with 3 components, only these 3 components are rotated. As you add components to the model, there is less room available for the PLSR rotation. As a result, as you add components to the model the PLSR coefficients get closer to PCR's. When you use all components, the PLSR and PCR coefficients are identical.

Also, the R^2 in the X row for all components is identical to the OLS R^2

Comp 19	PCR	PLSR
AtBat	-291.65	-291.65
Hits	338.47	338.47
HmRun	37.93	37.93
Runs	-60.69	-60.69
RBI	-27.05	-27.05
Walks	135.33	135.33
Years	-16.73	-16.73
CAtBat	-391.78	-391.78
CHits	86.85	86.85
CHmRun	-14.21	-14.21
CRuns	481.66	481.66
CRBI	261.19	261.19
CWalks	-214.30	-214.30
LeagueN	31.31	31.31
DivisionW	-58.53	-58.53
PutOuts	78.91	78.91
Assists	53.83	53.83
Errors	-22.20	-22.20
NewLeagueN	-12.37	-12.37

	19 comps
X	100.00
Salary	54.61

```
Multiple R-squared:  0.5461
```

Summary on Dimensionality

Quick Notes on Model Selection:

- If your analytics goal is predictive accuracy, select the model (either PCR or PLSR) and number of components that yield the lowest CV RMSE, misclassification or deviance.

- If you are interested in reducing dimensionality for either PCR or PLSR, select the smallest number of components that explain at least 70% of the variance of the predictors. Select either PCR or PLSR depending on which gives you the smallest number of components that meet this 70% criteria, or the one with the smallest CV RMSE.

- If you analytics goal is interpretation or inference, you want low bias in your model to the extent possible. Then, select the model with the largest number of components from either PCR or PLSR, with an acceptable CV RMSE

Summary for Models with Dimensionality Issues

- ✓ Variable Selection – use when: (1) exploring variables and adding/removing predictors is OK; and (2) when the goal is interpretation. Use Subset or Step methods to find the predictor set that maximizes predictive accuracy while keeping multicollinearity at tolerable levels.
- ✓ Shrinkage or Regularization – use when: (1) removing predictors is not an option. Ridge is better when all predictors must be included. LASSO is better when some variable coefficients can shrink to 0.
- ✓ Dimension Reduction Methods – use when: (1) the number of predictors is very large relative to the number of data observations available; (2) you must use all predictors; and (3) interpretation is NOT a goal, but predictive accuracy is. No preference between PCR and PLS → select the one with the lowest CV Test MSE or RMSE